Sensors and Sensor Technologies

Sensing Through the Lens of Musical Instruments

Romain Michon
Université Jean Monnet, Physical Interaction Design



"Musical instruments have the best interface" (Jaron Lanier)

Questions for today



https://www.youtube.com/watch?v=SYQ4yfWyYpM

- What physical events can an instrument sense?
- How do different sensor technologies shape the performer's actions?
- How does a sensor communicate its measurement to a microcontroller?
- What makes a sensing relationship playable and learnable?

A sensor produces information. Instrument design gives that information musical meaning.

What counts as sensing?

A sensor is a device or system that responds to a physical phenomenon and produces a signal that another system can use.

Example	Physical change	Possible observation
Potentiometer	Rotation	Changing voltage
Button		

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Microphone		

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Microphone	Air pressure	Electrical audio signal

A sensor can be a component, a module, or a larger sensing system.

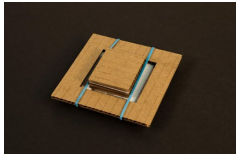
A sensor does not read intention

- A sensor turns a physical event into a signal.
- An accelerometer reports acceleration, not “shake the sound.”
- A camera can track a hand without knowing whether the performer is playing or resting.
- The same measurement can control brightness, pitch, spatial position, or nothing.
- Gesture meaning depends on the instrument, the performer, and the situation.

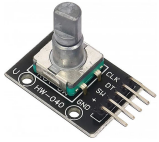
Discussion

How could a system distinguish an intentional gesture from an accidental bump?

Buttons and rotary controls



A DIY cardboard button.



An encoder.

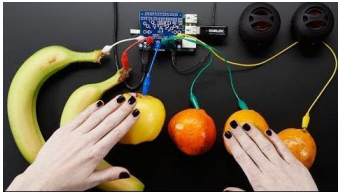
Switches and buttons

- Discrete state: pressed / released
- Clear onset and direct cause
- Good for triggering or selecting
- Can be hard to reach or trigger accidentally
- Many button technologies exist

Potentiometers and encoders

- Knob or slider movement
- Continuous value or stepped movement
- A physical position can show the setting
- Range, speed, and findability matter

Touch and pressure are different



Capacitive touch sensing.



Various FSRs.

Capacitive touch

- Detects finger presence or contact
- Can support broad touch surfaces
- Does not necessarily measure pressing force

Force and pressure

- Can provide a graded response to force
- Useful for dynamics, articulation, or density
- Consistency and tactile feedback matter

Examples: capacitive pads, force-sensitive resistors, load cells, pressure sensors.

Nuance: adding force sensitivity to capacitive touch sensing



https://youtu.be/r6IeF8eWvaI?si=XWAbG9u6gzSukKyc

Bending, stretching, and vibration

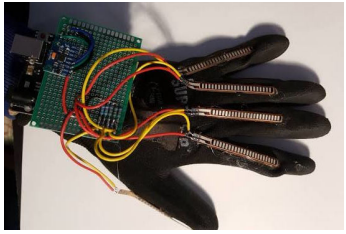


Flex sensors on a glove (left) and conductive textile-based controller (right)

- Flex sensors and conductive textiles respond to deformation.
- They can make a glove, garment, or soft object part of the interface.
- Piezoelectric elements respond to bending, impacts, or vibration.
- A piezo can reveal how an object is struck or resonating.

Design questions: Is it comfortable? Does it return to the same value? Does mounting change the response? What else might trigger it?

One gesture, several sensing choices



Suppose a performer bends a finger to change timbre.

Approach	What it measures	What it asks of the performer
Flex sensor	Finger bending	Wear or attach a sensor
Camera	Visible finger position	Stay in view; avoid occlusion
Hand IMU	Hand motion	Accept that finger motion is not isolated
Pressure pad	Squeezing or pressing	Change the gesture to contact pressure

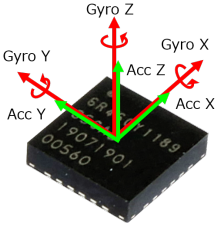
The sensing choice changes the action, not just the data format.

A short observation break

5 minutes

Find one object that senses something.
What would change if it became a musical interface?

Inertial sensors

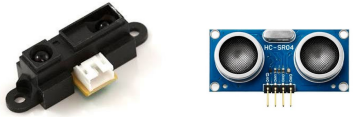


A 6dOf sensor.

- An accelerometer measures linear acceleration. Gravity can also help estimate tilt.
- A gyroscope measures rotational speed.
- An Inertial Measurement Unit (IMU) combines motion sensors, often across several axes.
- A magnetometer measures magnetic field direction.

Useful for: tilt, shake, rotation, movement intensity, handheld or wearable control.
Consider: sensor orientation, placement, drift, calibration, and motion ambiguity.

Proximity and distance



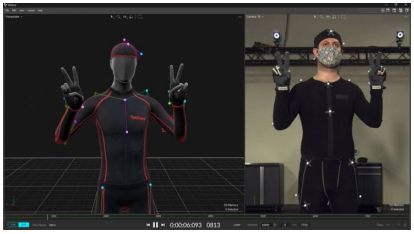
Infrared (left) and ultrasonic (right) distance sensors.

- Infrared, ultrasonic, time-of-flight, and radar sensors can detect nearby objects or distance.
- Contact-free sensing can make movement through space part of the instrument.
- The sensing region has a range and a field of view.
- Sunlight, reflective surfaces, occlusion, and nearby performers can affect detection.
- The lack of haptic feedback can be an issue.

Discussion

Does the performer have freedom to move anywhere, or only inside a designed sensing zone?

Cameras and computer vision



Optitrack system.

- A camera measures patterns of light in an image.
- Software can track markers, color regions, hands, body joints, or objects.
- Depth cameras add distance information to image regions.
- Lighting, background, viewpoint, and occlusion affect what can be detected.
- AI now plays a central role in this kind of sensing technologies.
- Tens of libraries exist in Max/MSP, PureData, TouchDesigner, etc. and on the Web for turning a camera into a sensor:
  `https://monicalim.online/Handmate-MIDI/`

Tracking a body part is different from identifying what the performer intends.

Microphones and acoustic sensing



Ge Wang playing Smule's Ocarina.

- A microphone converts air-pressure changes into an electrical signal.
- The signal can capture voice, breath, claps, taps, or an instrument.
- Sound itself can become the control gesture, without another surface.
- Background sound, room acoustics, loudspeakers, and feedback complicate detection.

Discussion

What could a microphone hear besides the intended performer?

Body and environmental signals



EEG sensors on Romain's head.

Physiological

- EMG: muscle activity
- ECG: heart activity
- EEG: electrical brain activity
- Placement and interpretation matter; these signals do not transparently reveal emotion or intention.

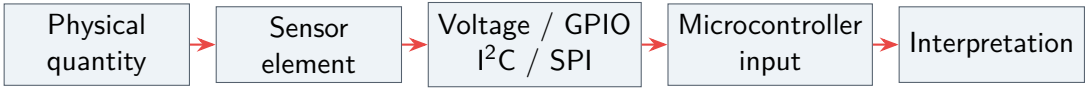


Weather station full of sensors.

Environmental

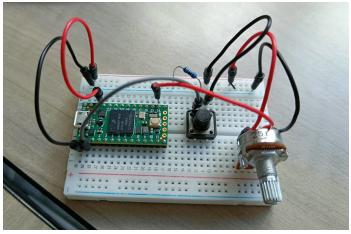
- Light, temperature, humidity, air quality
- Often slower than a performance gesture
- Can shape an evolving installation or soundscape
- The audience may not perceive the cause of change

Sensing principle and electrical interface



The sensor says what changes. The interface says how the measurement reaches the microcontroller.

Analog outputs and ADC inputs



A potentiometer connected to a Teensy.

- An analog sensor may provide a changing voltage.
- An ADC samples the voltage and represents it as a number.
- A potentiometer used as a voltage divider is a familiar example.
- A resistive element, such as a thermistor or force-sensitive resistor, needs a circuit to produce a readable voltage.

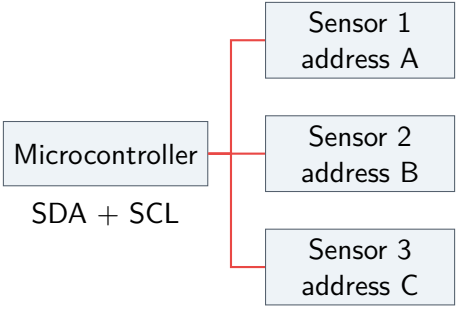
Check: input voltage limits, ADC range and resolution, number of channels, and any needed signal conditioning.

Digital inputs and pulse timing

- A digital GPIO input reads a logic state, often LOW or HIGH.
- Buttons need a stable default state, usually from a pull-up or pull-down.
- Contact bounce can make one press appear as several transitions.
- Some modules encode a reading as pulse duration or frequency; the microcontroller measures timing on a GPIO pin.

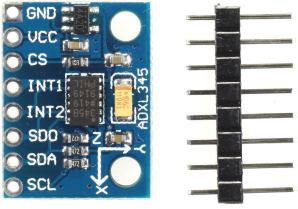
Digital GPIO is a logic-level connection. I^2C and SPI are serial data buses.

I²C: two shared signal wires



- SDA carries data; SCL carries the clock.
- Multiple addressed devices can share the bus.
- Check voltage compatibility, device addresses, wiring, and bus speed.
- The Arduino environment usually provides libraries for using “digital sensors.”

SPI: clocked data with device select



Digital accelerometer.

- Common signals: SCLK, MOSI, MISO, and a chip-select line.
- Often supports higher data rates than I^2C.
- Common for fast motion sensors, displays, and memory devices.
- More signal wires are needed; each device usually needs a chip-select line.
- Clock mode and maximum speed must match the sensor.

SPI can transfer data in both directions at once in common configurations.

UART and other connections

- Universal Asynchronous Receiver-Transmitter
- UART uses TX and RX lines without a shared clock.
- Both devices need matching data rate and format.
- Some sensor modules use UART; GPS modules are a familiar example.
- USB more often connects a microcontroller or interface to a computer than a bare sensing element to a pin.
- Other options include one-wire buses and CAN for particular systems.

Interface comparison: signal to pin

Output	Example	Main consideration
Analog voltage	Potentiometer	ADC range, voltage limits, channels
Resistance change	Thermistor, FSR	Needs divider or conditioning circuit
Digital GPIO	Button	Pull-up/down and contact bounce
Pulse timing	Ultrasonic module	Measure pulse width or frequency

Interface comparison: serial buses

Bus	Wires	Typical tradeoff
I^2C	SDA, SCL	Two signal wires; addressed devices share the bus
SPI	Clock, data in/out, chip select	Higher throughput; more wires and select lines
UART	TX, RX	Simple point-to-point serial link; settings must match

Every interface also needs compatible power, ground, and logic voltage.

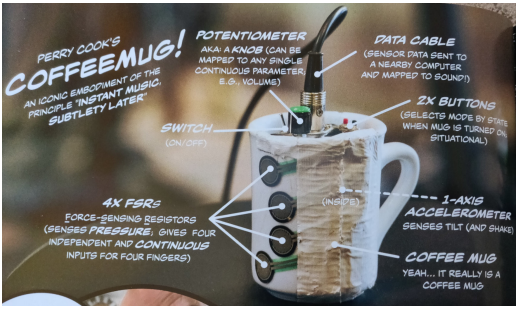
Three layers in one example: an accelerometer

1. **Physical quantity:** acceleration along sensor axes.
2. **Electrical interface:** readings arrive over I^2C or SPI.
3. **Musical interpretation:** motion intensity could control brightness, but the instrument defines that relationship.

Discussion

Would a digital interface make the motion reading more accurate or the instrument more expressive by itself?

Choosing a sensor starts with the interaction



Perry Cook's mug.

- What physical action should matter: touch, pressure, rotation, vibration, breath, or movement through space?
- What should the performer feel: contact, resistance, weight, freedom, or no object?
- Is the control a trigger, continuous value, position, direction, or gesture category?
- Where will it be used: desk, stage, gallery, or moving performance?
- What does the audience need to understand about cause and effect?
- Perry Cook's principles: `https://arxiv.org/pdf/2010.06524`

Four design tradeoffs

- **Contact and freedom:** wearables make sensing direct but can restrict movement; cameras allow free movement but need a clear view.
- **Precision and expressiveness:** stable values help accuracy; variation can support responsive sound.
- **Visible and hidden sensing:** visible controls clarify cause; hidden sensing can surprise but obscure the relationship.
- **Rich data and playable control:** more channels do not guarantee a better instrument.

More data does not guarantee more expression

Controller A

- One pressure value
- Clear connection to sound
- Repeatable and easy to learn

Controller B

- Camera tracks many body landmarks
- Many possible control values
- Relationship to sound is hard to predict

Expression depends on action, feedback, and learnability as well as the measured data.

The importance of haptic feedback



Nicolas Castagné and Claude Cadoz and their "Transducteurs gestuels rétroactifs".

- Haptic feedback simulates the sense of touch.
- It can be implemented using passive or active techniques.
- A simple rubber band can make a Theremin more playable.
- Active haptic feedback can be deployed with electrical, vibratory and mechanical technologies.
- For more on this topic, check out work by Sile O'Modhrain and Edgar Berdahl.

Group design activity

Choose one brief:

- A wearable controller changes timbre through a slow hand gesture.
- A wooden object responds differently to tapping, rubbing, squeezing, or tilting.
- A performer moves a sound source around an audience without touching a controller.
- Breath shapes a sustained sound and triggers discrete events.

Sketch the performer, sensor placement, and sound change.

Design sketch checklist

- What action should the performer make?
- What does the sensor actually measure?
- How does the output reach the microcontroller: ADC, GPIO, pulse timing, I^2C, or SPI?
- What musical event or parameter changes?
- What could fail or be confused with an intentional action?
- How will the performer know the system detected the action?